Publishing Android Applications

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Android app lifecycle

1. Idea and Planning
2. Design (UI/UX)
3. Development (coding)
4. Testing (Debugging/ QA)
5. Publishing (Deployment)
6. Maintenance and updates

What is app publishing?

- The process of making an app available to users
- Typically done via app stores
- Includes **preparation**, **submission** and **maintenance**

It is the final stage of the app lifecycle and includes updates and version management after release.

Preparation- Pre-publishing

- App readiness – remove debug code, test for crashes, optimize memory and battery
- Versioning – Select **Version Code** (Integer; increments every version) and **Version Name** (should be user-friendly)
- App signing - digitally signing an Android app with a private key to verify developer identity and ensure secure updates.

  Debug Signing– for testing only

  Release Signing– for publishing

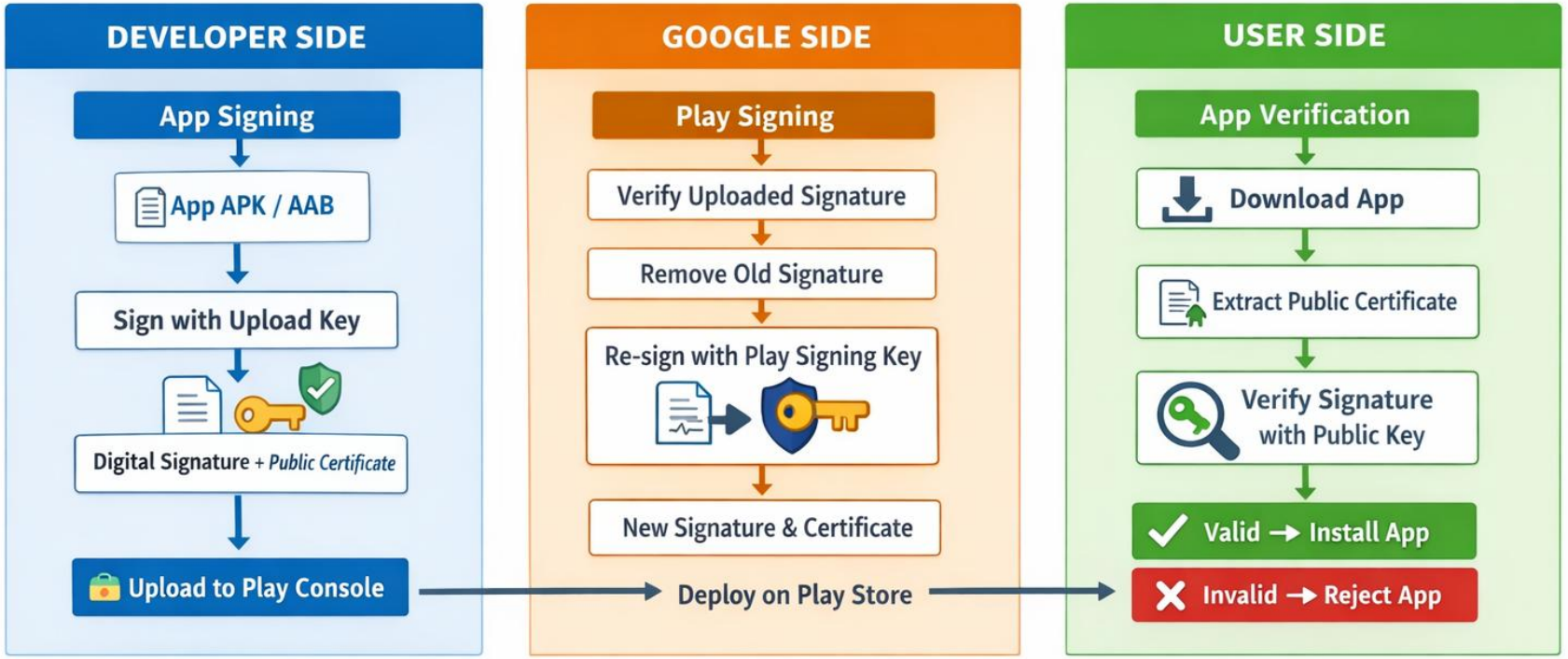
  Generate signed APK/AAB

  Keystore = critical; never lose it!

Preparation – Keystore in Release App Signing

Keystore contents:

Preparation – Release App Signing and Verification



Preparation – Debug App Signing and Verification

Preparation - Build and Packaging

- Build and packaging ensures **how your code becomes a distributable file** that can be uploaded to Google Play.
- Mainly two types of distributable files are used: APK (Android Package Kit) and AAB (Android App Bundle – introduced in 2018).

Preparation – Build and Packaging - APK vs AAB

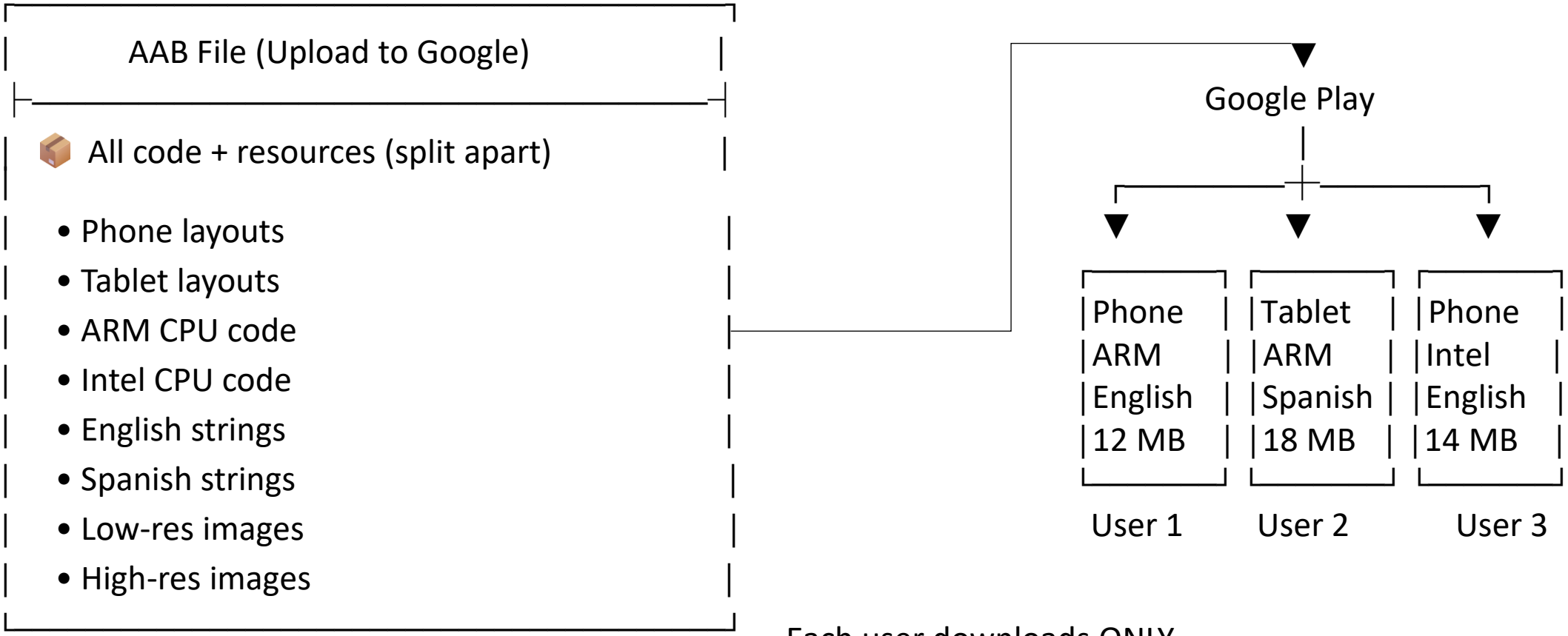
APK File

- Code for ALL devices
- Images for ALL screens
- Resources for ALL CPUs
- Languages for ALL regions

Total: 50 MB (example)

Problem: Every user downloads EVERYTHING, even if they don't need it.

Preparation - Build and Packaging - APK vs AAB



Each user downloads ONLY what their device needs!

Preparation - Build and Packaging - Generate Signed Bundle / APK

Creating a **signed (release) version** of your app that is ready for publishing on the Google Play Store.

In Android Studio:

1. Open the Option

Go to:
Build → Generate Signed Bundle / APK

2. Choose Format

Select:

Android App Bundle (AAB) → for Play Store
APK → for testing/manual install

Preparation - Build and Packaging - Generate Signed Bundle / APK (contd.)

Creating a **signed (release) version** of your app that is ready for publishing on the Google Play Store.

3. Select or Create Keystore

If you already have one → choose it

Else → click **Create New**

You'll enter:

Keystore location

Password

Key alias

Validity (e.g., 25 years)

Preparation - Build and Packaging - Generate Signed Bundle / APK (contd.)

4. Configure Build

Choose:

Build Variant → release

Select signature versions:

V1 (Jar Signature)
V2 (Full APK Signature) (recommended)

5. Generate File

Click **Finish**

Android Studio builds the signed file

Submission - Popular app distribution platforms

Deployment-ready app

Google play store

Third-party app stores

Direct APK distribution

Submission - The Play Store Ecosystem

It is an ecosystem managed by Google and it includes:

- Developers (app builders)
- Users (App downloaders)
- Policies (Rules that ensure safety and security)
- Monetization systems (ads, in-app purchases)

The central tool for developers is Google Play Console, where apps are uploaded, managed and monitored.

Submission - Creating developer account on Google play console

- **Fee**: One-time registration (~$25 USD, varies by country)
- **Account Types**:

Individual – for solo devs

Organization – for companies (requires legal documents)

Submission - App Listing

App listing is the **public page of your app** on the Google Play Store that users see before downloading it.

Asset	Details
Title	Short, memorable, includes keywords
Short Description	80 characters max
Long Description	4000 characters max
Keywords	For ASO (App Store Optimization)
App Icon	512×512 PNG
Screenshots	Phone + tablet (minimum 2–8 per device type)
Feature Graphic	1024×500 (appears in search results)
Promo Video	Optional YouTube link

Submission – Policies and Compliance

Google Play policies are a set of **rules and guidelines** defined by the Google Play Store that developers must follow to publish and maintain apps.

These policies ensure:

- User safety
- Data privacy
- High-quality apps

Submission – Policies and Compliance (Contd.)

Requirement	Notes
Google Play Policies	Read – especially privacy, content, ads, and impersonation
Permissions	Request only what you need; explain why
Data Safety Section	Disclose all data collection/sharing
Privacy Policy	Required if app collects personal data → host a public URL

Submission – Steps to upload the app

1. Create new app in Play Console
2. Upload .aab file
3. Add release notes (what's new)
4. Choose **release track**:

Track	Purpose
Internal	Quick tests (up to 100 testers)
Closed	Trusted beta group (limited users)
Open	Public beta (anyone can join)
Production	Live for all users

Submission – Publishing the app

- After submitting to **Production**, Google reviews your app
- Review time: hours to several days (varies)
- **Common rejections**:
  - Policy violation (e.g., misleading ads)
  - App crashes on review
  - Incomplete data safety form
  - Broken privacy policy link

Maintenance - Post-Publishing Management

Activities performed after an app is released on the Google Play Store. It involves maintaining, improving, and monitoring the app over time.

Activity	Reason
Update app	Increment version code, upload new AAB, repeat review
Monitor reviews/ratings	Respond to users, improve rating
Analytics	Track installs, uninstalls, retention (Play Console + Firebase)
User feedback	Fix bugs, add requested features

Thank you